

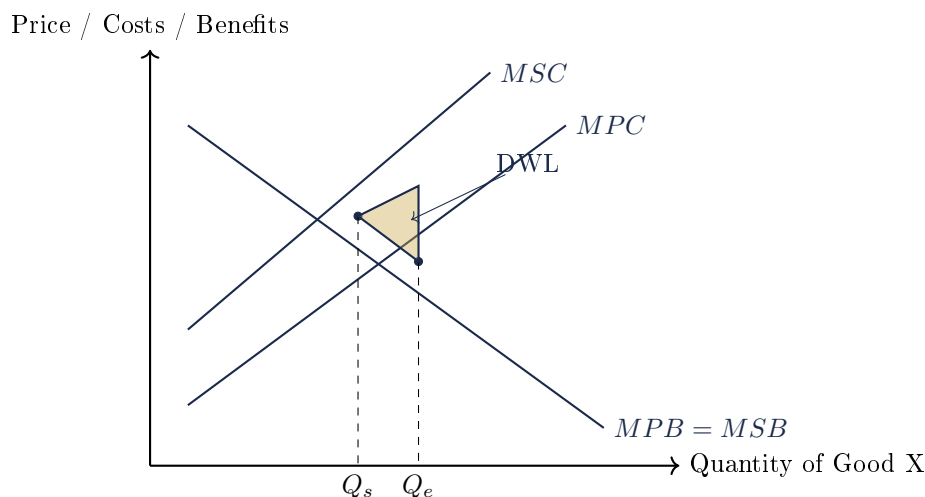
Document 7. Market Failure: Externalities

Topic overview. An externality is a spillover cost or benefit imposed on third parties not party to the transaction. The rational consumer or producer maximises their private benefit / cost, ignoring third-party effects, so the free market over- or under-allocates resources. The result is a deadweight welfare loss relative to the social optimum.

Section A. Important Definitions

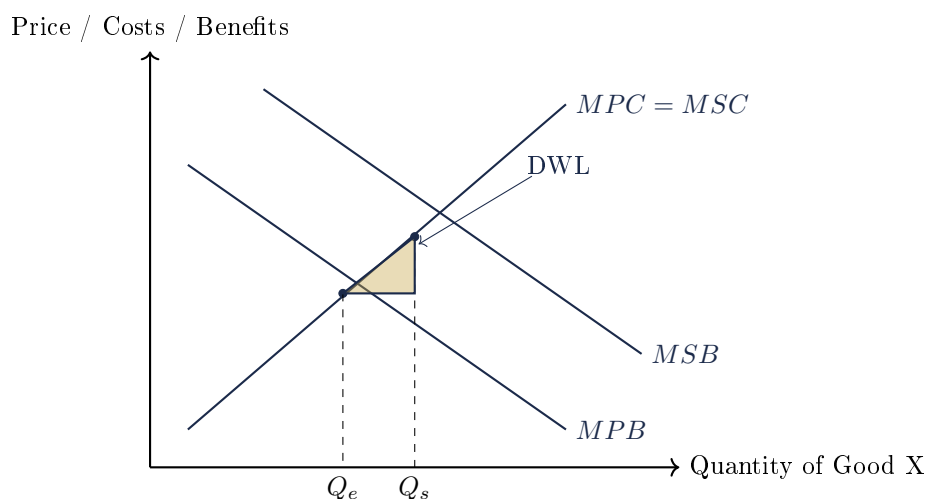
Externality	A spillover cost or benefit to third parties who are not directly involved in the production or consumption of goods and services.		
Marginal Private Cost (MPC)	Additional cost incurred by producers / consumers from one more unit.		
Marginal External Cost (MEC)	Additional cost imposed on third parties from one more unit.		
Marginal Social Cost (MSC)	$MSC = MPC + MEC$. Total opportunity cost to society.		
Marginal Private Benefit (MPB)	Benefit obtained by producers / consumers from one more unit.		
Marginal External Benefit (MEB)	Benefit enjoyed by third parties from one more unit.		
Marginal Social Benefit (MSB)	$MSB = MPB + MEB$. Total benefit to society.		
Allocative efficiency	Achieved when $MSB = MSC$ (society's welfare is maximised).		
Deadweight Loss (DWL)	Loss of total welfare from over- or under-allocation of resources.		

Section B. Diagram 1 — Negative Externality in Consumption (e.g. cigarettes)



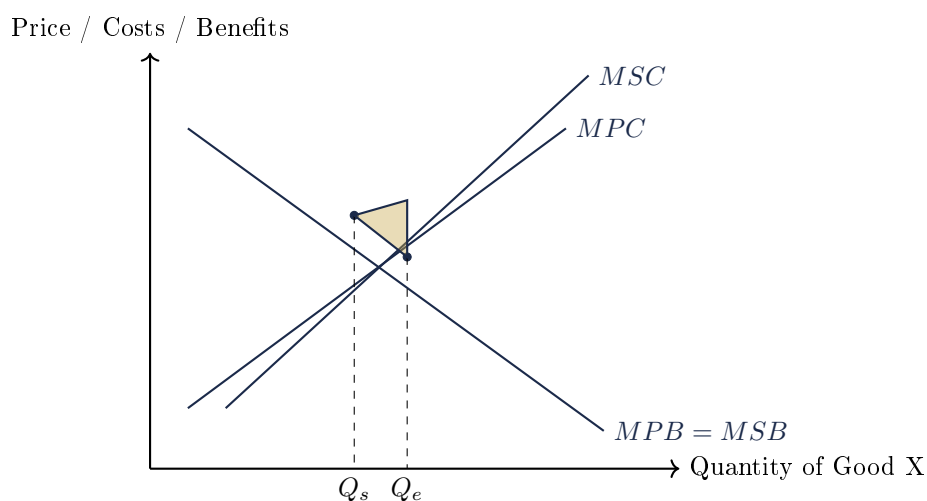
Caption. Free market equilibrium Q_e where $MPB = MPC$. Socially optimal output Q_s where $MSB = MSC$. Since $Q_e > Q_s$, there is *over-consumption* of cigarettes. The shaded triangle is the deadweight loss.

Section C. Diagram 2 — Positive Externality in Consumption (e.g. vaccination, education)

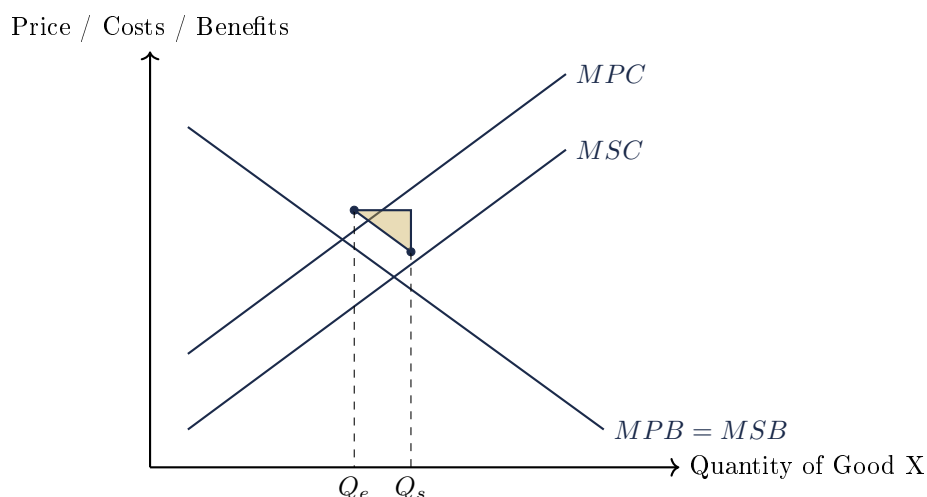


Caption. Free market equilibrium Q_e where $MPB = MSC$. Socially optimal output Q_s where $MSB = MSC$. Since $Q_e < Q_s$, there is *under-consumption*.

Section D. Diagram 3 — Negative Externality in Production (e.g. factory pollution)



Section E. Diagram 4 — Positive Externality in Production (e.g. R&D spillovers)



Section F. Question Type 1: Explain how a [negative / positive] externality leads to market failure

Technique: apply the 7-step framework below verbatim. Substitute the underlined slots with the specific externality (good X), the responsible party, and the third-party effect.

Step 1. Assuming that there are no negative / positive externalities in the market for (good X), the marginal social cost / benefit is equal to the marginal private cost / benefit.

Step 2. The free market equilibrium is at Q_e units, where $MPB = MPC$. Consumers and producers only consider their private benefits and costs when making decisions. The private benefits include (insert example), while the private costs include (insert example).

Step 3. However, at Q_e , there is a Marginal External Cost / Benefit represented by the distance between the MSC / MPC / MSB / MPB curve and the MSC / MPC / MSB / MPB curve because the (private economic agent) ignores the positive / negative externalities created from the (activity).

Step 4. The overall benefit to society is reduced / higher because third parties who (explain external cost / benefit) incur additional costs / enjoy additional benefits. Hence, output Q_e is allocatively inefficient.

Step 5. The socially efficient output is instead at Q_s , where $MSC = MSB$. Thus, there is an over- / under-production / consumption of (good X) by $Q_e Q_s$ units.

Step 6. This results in a deadweight loss to society represented by the shaded triangle.

Step 7. As too much / little scarce resources are allocated to the production and consumption of (good X), the market fails to achieve allocative efficiency, resulting in market failure.

NOTE: Framework reproduced word-for-word from the Crucible Education Centre handout to preserve calibrated language.

Section G. Framework for Memorising the Diagram

Step 1. Draw an X. Label the MPC and MPB. Label Q_e .

Step 2a (positive externality). Benefits must diverge. Keep costs constant: label $MPC = MSC$.

Step 2b (negative externality). Costs must diverge. Keep benefits constant: label $MPB = MSB$.

Step 3a (positive externality). The government wants more consumption / production, so draw MSB such that Q_s is on the *right* of Q_e . Label Q_s and MSB.

Step 3b (negative externality). The government wants less consumption / production, so draw MSC such that Q_s is on the *left* of Q_e . Label Q_s and MSC.

Step 4. Shade the deadweight loss triangle. Remember that it must point towards Q_s .

Section H. Question Type 2: Evaluate government policies to correct externalities

Policy	How it works	Strengths / weaknesses
Pigouvian tax	Tax = MEC; shifts MPC up to MSC.	Internalises externality; raises revenue. <i>Weakness</i> : requires accurate MEC measurement; regressive; depends on PED.
Subsidy	Subsidy = MEB; shifts MPC down or DD up.	Encourages consumption. <i>Weakness</i> : fiscal cost; deadweight loss elsewhere.
Tradable permits	Cap and trade pollution rights.	Market-driven; cost-minimising. <i>Weakness</i> : cap-setting; political pressure; carbon leakage.
Regulation / ban	Quota, emissions standard.	Direct, fast. <i>Weakness</i> : no incentive to over-comply; enforcement cost; black markets.
Direct provision	State provides the good.	Universal access; equity. <i>Weakness</i> : crowding-out; X-inefficiency.
Education campaigns	Shift MPB toward MSB.	Long-run cultural change. <i>Weakness</i> : slow; uncertain magnitude.

Section I. Singapore Applications

- **Carbon tax** (S\$25/tCO₂e in 2024, rising to S\$45 by 2026–27).
- **COE quota** and **ERP** address congestion externalities.
- **Tobacco / alcohol excise taxes**; minimum legal age laws; pictorial health warnings.
- **Mandatory schooling** and **MOE subsidies** for positive education externalities.
- **Subsidised MediShield Life** and **childhood vaccinations** address under-consumption of preventive healthcare.

Section J. Worked Examples

Worked Example 1: Singapore carbon tax

Q. Using a diagram, explain how Singapore's carbon tax corrects the negative externality of carbon emissions.

A Pigouvian tax = MEC raises producers' MPC up to MSC. New private equilibrium coincides with the social optimum Q_s . Output falls from Q_e to Q_s , eliminating the DWL. Revenue is recycled into clean-energy subsidies. *Evaluation*: the tax rate may still be below true MEC (IPCC estimates US\$50–150/tCO₂e); small emitters and households are exempt; firm relocation (carbon leakage) is a risk.

Worked Example 2: Vaccine subsidy

Q. Evaluate the use of subsidies to correct under-vaccination.

Subsidy lowers the effective price, raising private consumption toward Q_s . Equity-enhancing for low-income households. *Limitations*: fiscal cost; if PED is low, subsidy mostly raises producer surplus; behavioural barriers (vaccine hesitancy) may dominate price effects. Mandate or campaign may be needed jointly.